

REAL ESTATE LISTING PORTAL

A Project Synopsis Report Submitted To



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PROJECT SYNOPSIS

1. Project Title

Real Estate Listing Portal with Interactive Maps, Virtual Tours, and Agent Communication System

Tagline: *"From Search to Settlement – Everything in One Place."*

2. Abstract

The Real Estate Listing Portal is a web-based application designed to simplify the process of buying, selling, and renting properties. The platform allows users to search for properties using advanced filters such as location, price, property type, and amenities. Integration with Google Maps provides interactive map-based search, nearby facilities, and location insights.

The system supports virtual property tours, enabling users to explore properties remotely before making a visit. A mortgage calculator helps buyers estimate monthly payments and financial commitments. Real-time chat functionality allows direct communication between buyers and agents, improving transparency and customer engagement.

The application is developed using React.js for the frontend, Node.js and Express.js for the backend, PostgreSQL with PostGIS for geospatial property searches, and MongoDB for chat message storage. Additional features such as saved searches, property price history, comparable sales, payment gateway integration, and mobile responsiveness enhance the overall user experience.

The proposed system provides a modern, efficient, and technology-driven solution for real estate transactions, making property discovery and decision-making faster, easier, and more convenient.

3. Problem Statement

Finding and purchasing properties is often time-consuming because users need to visit multiple websites, contact different agents, and physically inspect properties. The proposed system provides a centralized platform for property search, virtual tours, map-based navigation, and agent communication.

4. AIM

The aim of this project is to develop a smart and user-friendly Real Estate Listing Portal that enables buyers to search properties efficiently using interactive maps, advanced filters, virtual tours, and real-time communication with agents. The system aims to simplify the property buying and selling process by providing all essential services on a single platform.

5. Objectives

- 1) To build a Zillow-style property search portal with advanced search filters.
- 2) To integrate Google Maps API for location-based property search and nearby amenities.
- 3) To provide virtual property tours for remote viewing of properties.

- 4) To implement a mortgage calculator for estimating loan payments.
- 5) To enable real-time communication between buyers and agents through chat.
- 6) To provide saved searches and listing alerts for users.
- 7) To support secure online payments for premium listings and booking deposits.
- 8) To develop a responsive web application accessible on desktop and mobile devices.

6. Scope

The scope of the Real Estate Listing Portal is to provide a centralized platform for buyers, sellers, and real estate agents to search, list, and manage properties online. The system allows users to search properties using advanced filters and interactive maps, view virtual tours, calculate mortgage payments, and communicate directly with agents. It helps reduce manual efforts, saves time, and improves transparency in property transactions. The platform can be used by individuals, real estate agencies, and property dealers and can be accessed through desktop and mobile devices. Future enhancements such as AI-based recommendations, property price prediction, and AR/VR property tours can further improve the system.

7. Methodology

The proposed Real Estate Listing Portal follows the Software Development Life Cycle (SDLC) methodology for systematic development and implementation. The project is developed through the following phases:

- 1) **Requirement Analysis :-** In this phase, the requirements of buyers, sellers, and real estate agents are collected and analyzed. Features such as property search, map integration, virtual tours, mortgage calculation, and real-time chat are identified.
- 2) **System Design :-** The overall system architecture, database structure, user interface design, Use Case Diagram, and Data Flow Diagram (DFD) are prepared to define the workflow of the application.
- 3) **Development :-** The application is developed using React.js for the frontend, Node.js and Express.js for the backend, PostgreSQL with PostGIS for property data management, and MongoDB for chat message storage.
- 4) **Testing :-** The developed system is tested to verify the functionality of property search, map-based navigation, virtual tours, chat system, and payment features. Any identified errors are corrected.
- 5) **Deployment :-** After successful testing, the application is deployed and made available for users to search properties, communicate with agents, and perform property-related activities online.

8. Process Description / System Workflow

The Real Estate Listing Portal works through a series of steps that allow buyers and agents to interact efficiently.

- 1) The user registers and logs into the system.
- 2) Agents add and manage property listings by uploading property details, images, prices, and location information.
- 3) Property information is stored in the database.

- 4) Buyers search for properties using filters such as location, price, property type, and amenities.
- 5) The system retrieves and displays matching properties on an interactive Google Map.
- 6) Buyers can view property details, images, floor plans, and virtual tours.
- 7) The mortgage calculator helps users estimate monthly loan payments based on property price and loan terms.
- 8) Buyers can communicate directly with agents using the real-time chat system.
- 9) Users can save searches and receive notifications for matching properties.
- 10) Agents can manage listings, respond to buyer queries, and monitor property performance.
- 11) The system processes payments for premium listings and booking deposits.
- 12) The administrator manages users, verifies listings, and generates reports.

9. Use Case Diagram

Buyer	Agent	Admin
• Register/Login • Search Property • View Property Details • Take Virtual Tour • Use Mortgage Calculator • Chat with Agent • Save Searches • Book Property	• Register/Login • Add Property • Manage Listings • Chat with Buyers • View Analytics	• Manage Users • Verify Listings • Manage Payments • Generate Reports

10. Hardware and Software Requirements

Hardware Requirements :- The proposed system requires a computer with an Intel Core i5 processor or higher, 8 GB RAM, 256 GB SSD storage, and an internet connection

Software Requirements :- The project will be developed using React.js, HTML5, CSS3, and JavaScript for the frontend,

while Node.js and Express.js will be used for the backend.

PostgreSQL with PostGIS and MongoDB will be used as databases.

Google Maps API, Socket.io, Matterport/Kuula, and Stripe Payment Gateway will be integrated to provide map-based search, real-time chat, virtual tours, and online payment facilities.

Development and testing will be carried out using Visual Studio Code, GitHub, Postman, and Google Chrome.

11. Future Scope

- AI-based property recommendation system for personalized property suggestions.
- Property price prediction using Machine Learning techniques.
- AR/VR-enabled virtual property tours for an immersive viewing experience.
- AI chatbot integration for instant customer support and query resolution.
- Development of Android and iOS mobile applications.
- Blockchain-based property verification for secure and transparent transactions.
- Smart notification system for new property listings and price updates.

12. Project Schedule (Gantt Chart)

TASK / ACTIVITY	WEEK-1 (Day 1 - 7)	WEEK-2 (Day 8 - 14)	WEEK-1 (Day 15 - 21)	WEEK-1 (Day 22 - 30)
1. Requirement Analysis	✓			
2. System Design	✓	✓		
3. Database Design	✓	✓		
4. Frontend Development		✓	✓	
5. Backend Development		✓	✓	
6. Google Maps Integration			✓	
7. Virtual Tour Integration			✓	
8. Chat System & Mortgage Calculator			✓	
9. Testing & Bug Fixing				✓
10. Deployment & Documentation				✓

13. Conclusion

The Real Estate Listing Portal simplifies property searching by combining maps, virtual tours, mortgage calculations, and agent communication into a single platform, making the property buying process faster and more convenient.

14. References

Books

1. Ian Sommerville – Software Engineering
2. Elmasri & Navathe – Fundamentals of Database Systems

Websites

1. React Documentation
2. Node.js Documentation
3. PostgreSQL Documentation & Google Maps API Documentation